Week 7 reading guide and worked examples

Weekly overview

Required learning outcomes

By the end of this week, you should be able to:

- solve equality-constrained optimisation problems with Lagrange multipliers, interpret a multiplier under the stated constraint encoding, and apply the envelope theorem (`M10`);
- formulate and solve elementary inequality-constrained optimisation problems using the course Kuhn–Tucker convention, complementary slackness and explicit interior and corner checks (`M11`); and
- distinguish parameters, estimands, estimators and estimates; evaluate elementary estimators by bias, variance, mean squared error and consistency; and derive simple method-of-moments and maximum-likelihood estimators (`S7`).

Prerequisites and retrieval

You will use curvature and sufficiency conditions from `M8`, unconstrained optimisation from `M9`, moments from `S4`, and sampling results from `S6`. Before beginning, retrieve the column-gradient convention, the negative-definite Hessian test, the distinction between a candidate and a global solution, and $\mathrm{Var}(\bar{X}) = \mathrm{Var}(X_1)/n$ for an iid sample with finite variance.

Question 10 contains three explicit cumulative retrieval parts on `M8`, `M9` and `S6`. Complete that block without consulting earlier notes.

Course convention for inequality constraints

For every maximisation problem, write inequality constraints as $g_j(\mathbf{x}) \geq 0$ and use

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = f(\mathbf{x}) + \sum_j \lambda_j g_j(\mathbf{x}) + \sum_k \mu_k h_k(\mathbf{x}), \qquad \lambda_j \geq 0.$$

The required conditions are stationarity, primal feasibility, dual feasibility and complementary slackness. Write non-negativity restrictions such as $x_i \geq 0$ as explicit constraints.

Student materials

Before the estimator and method-of-moments work, use the required estimators and method of moments bridge note. It is point-of-need support within the existing reading and worked-example allocation, not additional workload.

1. Follow the exact external sections and skip instructions in the reading guide.
2. Study the original institutional explanations and reproduce all four worked examples from a blank page.
3. Attempt every part of the exercise sheet without opening the full solutions.
4. Use the progressive hints only after a serious attempt, one stage at a time.
5. Mark your work from the complete solutions, record the first cause of each error by outcome ID, and reattempt missed parts two or three days later.
6. Complete the closed-note self-test, mark it immediately and update your error log.
7. Sit Checkpoint B under its separate timed-paper instructions.

The external pages remain on their authors' or institutions' sites. Open only the assigned sections. No video, applet, code or specialist software is required.

Suggested study sequence

1. Write the course Kuhn–Tucker convention at the top of your notes.

2. Review equality constraints first. For each multiplier interpretation, write the constraint encoding before deciding its sign.
3. Study the envelope theorem only after you can solve the corresponding constrained problem and state the regularity conditions.
4. For inequality constraints, list all four groups of Kuhn–Tucker conditions before choosing active and slack constraints.
5. Make a separate terminology card for parameter, estimand, estimator and estimate, then derive bias, variance and mean squared error from their definitions.
6. Derive method-of-moments and maximum-likelihood estimators on paper. State the parameter space and check boundary cases.
7. Attempt the weekly sheet, use hints only as needed, mark it and complete the retrieval block from memory.
8. Take the self-test, update the error log, then sit Checkpoint B.

Workload and required outputs

The times below are provisional planning guidance, not a guarantee, and have not yet been verified by students. They are subject to first-cohort calibration.

Component	Target time	Required output
Exact external reading and closed-source summaries	3 h 45 min	Original notes covering every stated condition, convention and estimator definition
Institutional worked-example study and reproduction	1 h 5 min	Four examples reproduced from a blank page
Week 7 exercise sheet	4 h	All 17 scorable parts attempted and marked
Self-test, marking and error-log update	30 min	Four self-test parts marked and outcome-labelled error entries updated
Checkpoint B	1 h 15 min	Separate timed attempt under checkpoint conditions
Core total	**10 h 35 min**	Completed Week 7 study and assessment cycle

The 225-minute external-reading estimate and 65-minute worked-example allocation give a 290-minute reading-and-example block. The resulting Week 7 total is within the general 9–11-hour planning range. All timings remain provisional planning guidance, not a guarantee, and are subject to first-cohort calibration.

The exercise sheet contains 10 numbered questions, 17 scorable parts, 100 marks and a 240-minute target. Across marks, it contains 30% foundation, 60% standard and 10% stretch work. The timings remain provisional planning guidance, not a guarantee, and are subject to first-cohort calibration.

Checkpoint B's timed sitting is included in the core total. Its required post-attempt self-marking, diagnosis and spaced reattempt are separate provisional activities: no student-validated minute allowance is claimed, and they are excluded from the published core total pending first-cohort calibration.

By the end of the week, keep:

- your constraint-convention note and estimator terminology card;
- four reproduced worked examples;
- a complete, marked first attempt at all 17 weekly parts;
- three closed-note retrieval attempts labelled `M8`, `M9` and `S6`;
- a marked self-test and a dated reattempt plan; and
- your separate Checkpoint B attempt.

End-of-week check

Do not move on merely because you can solve a first-order system. Check that you can explain why the candidate is feasible, why the stated sufficiency condition applies, how the constraint encoding determines a multiplier's interpretation, and why an estimator property concerns repeated sampling rather than one observed estimate.

Required reading route

How to use this route

Complete the four required written selections below before attempting the exercise sheet. Open only the exact sections listed and follow every skip and notation-conversion instruction. Allow about 225 minutes for active reading and brief closed-source summaries. The separate 65-minute allocation in worked_examples.md completes the combined 4-hour 50-minute reading and example-reproduction block. This is a planning estimate for the exact four-route package and remains approximate and student- unverified.

These four exact Week 7 routes, locators and skip instructions were approved on 1 August 2026 for external-link-only use. The course does not reproduce or adapt an external explanation, example, question or solution. If an external page's mathematical presentation is inaccessible to you, stop using that page, use the original institutional explanations and examples in this package, and report the access problem. To report a broken, inaccessible or unavailable required source, email cristiano.cantore@uniroma1.it with the week, item or source locator. You may request this confidentially without disclosing disability or other sensitive personal information. The programme Coordinator will provide an accessible replacement PDF or an approved alternative reading.

Required bridge note

Before the estimator and method-of-moments selections, read the estimators and method of moments bridge note. It standardises the course terminology within the existing Week 7 reading and worked-example allocation, not additional workload.

1. Equality constraints and the envelope theorem (`M10`)

Resource `MATH-M10-OSB-REQ`; source `OSBORNE_MM`. In Martin J. Osborne, *Mathematical Methods for Economic Theory*, read:

- Section 6.1.1;
- Section 6.1.2;
- Section 6.1.3;
- Section 6.1.4;
- in Section 6.2, the statement and one simple example only; and
- Section 6.3, "The envelope theorem".

Skip: general proofs in Section 6.2, advanced multiple-constraint examples, and exercises requiring results beyond `M8`–`M9`. Re-encode every equality constraint as $h_k(\mathbf{x}, \mathbf{a}) = 0$ and use $\mathcal{L} = f + \sum_k \mu_k h_k$. Derive a multiplier's sign and interpretation from that encoding.

Target reading and summary time: 65 minutes.

2. Elementary Kuhn–Tucker conditions (`M11`)

Resource `MATH-M11-OSB-REQ`; source `OSBORNE_MM`. In Osborne, read:

- Section 7.1, "The Kuhn–Tucker conditions";
- the sufficiency result and examples only in Section 7.3;
- Section 7.4, "Nonnegativity conditions"; and
- the Section 7.5 summary.

Skip: Section 7.2 in full, proofs in Section 7.3 and advanced constraint qualifications. The source uses a different sign convention. Translate every problem to the course convention: maximise with $g_j(\mathbf{x}) \geq 0$, use $\mathcal{L} = f + \sum_j \lambda_j g_j$, and require $\lambda_j \geq 0$. Write variable non-negativity as an explicit constraint.

Target reading and summary time: 55 minutes.

3. Estimators and maximum likelihood (`S7`)

Resource `STAT-S7-REQ-PISHRO`; source `PISHRO_NIK`. In Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes*, read the definitions of sample, statistic, estimator and estimate in §8.1.1, "Random Sampling", then read the following non-contiguous point-estimation sections in order:

- §8.2.0, "Point Estimation";
- §8.2.1, "Evaluating Estimators";
- §8.2.2, "Point Estimators for Mean and Variance"; and
- §8.2.3, "Maximum Likelihood Estimation".

Skip: Section 8.2.4 on asymptotic properties of maximum-likelihood estimators, Section 8.5 on linear regression, order statistics and all software-based work. Use the institutional terminology note in worked_examples.md for the additional distinction between a parameter and an estimand.

Target reading and summary time: 85 minutes.

4. Method of moments (`S7`)

Resource `STAT-S7-REQ-PSU-MOM`; source `PSU_STAT415_MOM`. In Penn State STAT 415, whose primary course author is Laura Simon, read Lesson 3, Section 3.2, "Method of Moments": the definitions, "One Form of the Method", and Examples 3.8 and 3.9 only.

Skip: "Another Form of the Method" and Examples 3.10 and 3.11, including gamma-distribution material. Do not navigate from this page into unassigned statistics topics or software work.

Target reading and summary time: 20 minutes.

Reading output

After each selection, close the page and write a brief original summary. By the end of the route, your notes should contain:

- the equality-constrained Lagrangian and the conditions needed for a candidate and a global conclusion;
- the envelope derivative under an explicitly stated constraint encoding;
- the complete course Kuhn–Tucker conditions, including explicit non-negativity constraints;
- definitions of parameter, estimand, estimator and observed estimate;
- the formula $\mathrm{MSE}(\widehat{\theta}) = \mathrm{Var}(\widehat{\theta}) + \mathrm{Bias}(\widehat{\theta})^2$;
- the population-moment equation used by the method of moments; and
- the likelihood, log-likelihood, parameter space and boundary checks used in elementary maximum likelihood.

Do not copy external prose, examples, questions or solutions into shared notes.

Worked examples

These original institutional explanations and examples support `M10`, `M11` and `S7`. They do not reproduce or adapt an external example. Allow about 65 minutes to study and reproduce them.

For each example, cover the **Solution**, follow the **Method**, then reproduce the derivation from a blank page. Do not memorise a multiplier or estimator without its conditions.

Equality constraints and multipliers (M10)

Consider the maximisation problem

$$\max_{\mathbf{x}\in\mathbb{R}^n} f(\mathbf{x},\mathbf{a}) \quad \text{subject to} \quad h_k(\mathbf{x},\mathbf{a})=0, \qquad k=1,\dots,m.$$

Under the course convention, the Lagrangian is

$$\mathcal{L}(\mathbf{x},\mu;\mathbf{a}) = f(\mathbf{x},\mathbf{a}) + \sum_{k=1}^{m} \mu_k h_k(\mathbf{x},\mathbf{a}).$$

At a regular local optimum, the equality-constraint gradients must have the required linear independence, and the first-order conditions are

$$\nabla_{\mathbf{x}}\mathcal{L}(\mathbf{x},\mu;\mathbf{a}) = \mathbf{0}, \qquad h_k(\mathbf{x},\mathbf{a})=0.$$

These conditions identify candidates. A separate argument is needed for a maximum. For example, if f is strictly concave and the equality constraints are affine, a feasible first-order candidate is the unique global maximiser.

A multiplier's sign has no interpretation until the constraint encoding is stated. If a scalar resource parameter a enters as $h(\mathbf{x},a)=a-r(\mathbf{x})=0$, increasing a changes the required equality level. It relaxes a resource constraint only when this equality is the binding form of an associated restriction $r(\mathbf{x})\le a$. Under suitable regularity and differentiability conditions, if a enters the problem only through this constraint (so $f_a=0$), the envelope theorem gives

$$\frac{\mathrm{d}V}{\mathrm{d}a} = \left.\frac{\partial\mathcal{L}}{\partial a}\right|_{(\mathbf{x}^*(a),\mu^*(a))} = \mu^*(a).$$

If the same feasible set were encoded as $r(\mathbf{x})-a=0$, the multiplier would have the opposite sign.

Example 1 — a nonlinear equality constraint and envelope derivative (M10)

For $a>0$, solve

$$\max_{x,y}\{x+y\} \quad \text{subject to} \quad x^2+2y^2=a.$$

Find the value function and verify the envelope derivative.

Method. Encode the constraint as $h(x,y,a)=a-x^2-2y^2=0$, solve the regular first-order system, compare all candidates on the compact feasible ellipse, and then apply the envelope theorem.

Solution. The Lagrangian is

$$\mathcal{L}(x,y,\mu;a) = x+y+\mu(a-x^2-2y^2).$$

The first-order and feasibility conditions are

$$1-2\mu x=0, \qquad 1-4\mu y=0, \qquad a-x^2-2y^2=0.$$

The first two conditions imply $x=2y$. Feasibility then gives the two candidates

$$(x, y) = \left(2\sqrt{\frac{a}{6}}, \sqrt{\frac{a}{6}}\right) \quad \text{and} \quad (x, y) = \left(-2\sqrt{\frac{a}{6}}, -\sqrt{\frac{a}{6}}\right).$$

The constraint gradient is non-zero everywhere on the ellipse because $a > 0$. The feasible set is compact, so a global maximum exists, and the two candidate objective values are respectively $3\sqrt{a/6}$ and $-3\sqrt{a/6}$. Therefore

$$x^*(a) = 2\sqrt{\frac{a}{6}}, \qquad y^*(a) = \sqrt{\frac{a}{6}}, \qquad \mu^*(a) = \frac{\sqrt{6}}{4\sqrt{a}}.$$

The value function is

$$V(a) = 3\sqrt{\frac{a}{6}} = \sqrt{\frac{3a}{2}},$$

so direct differentiation gives

$$V'(a) = \frac{\sqrt{6}}{4\sqrt{a}}.$$

The envelope calculation agrees:

$$\left.\frac{\partial \mathcal{L}}{\partial a}\right|_* = \mu^*(a) = \frac{\sqrt{6}}{4\sqrt{a}}.$$

Interpretation. Because the constraint was written $a - x^2 - 2y^2 = 0$, the positive multiplier measures the marginal value of increasing the quadratic-capacity level a. It is a relaxation only when the equality is the binding form of the associated capacity restriction $x^2 + 2y^2 \leq a$. Reversing the constraint encoding reverses the multiplier sign but leaves $V'(a)$ unchanged.

Inequality constraints and the course convention (`M11`)

The course-wide convention is a maximisation problem with

$$g_j(\mathbf{x}) \geq 0, \qquad h_k(\mathbf{x}) = 0.$$

Use

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = f(\mathbf{x}) + \sum_j \lambda_j g_j(\mathbf{x}) + \sum_k \mu_k h_k(\mathbf{x}), \qquad \lambda_j \geq 0.$$

The Kuhn–Tucker conditions are:

1. **Stationarity**

$$\nabla f(\mathbf{x}) + \sum_j \lambda_j \nabla g_j(\mathbf{x}) + \sum_k \mu_k \nabla h_k(\mathbf{x}) = \mathbf{0}.$$

2. **Primal feasibility**

$$g_j(\mathbf{x}) \geq 0, \qquad h_k(\mathbf{x}) = 0.$$

3. **Dual feasibility**

$$\lambda_j \geq 0.$$

4. **Complementary slackness**

$$\lambda_j g_j(\mathbf{x}) = 0.$$

Complementary slackness says that a slack constraint has a zero multiplier. It does not say that every zero multiplier corresponds to a slack constraint. For the elementary problems in this week, strict concavity of the objective and affine constraints provide a direct sufficiency route once a feasible Kuhn–Tucker point has been found.

Example 2 — a two-variable lower-bound corner (`M11`)

Solve

$$\max_{x,y} \; -(x+2)^2 - 2(y-1)^2 \quad \text{subject to} \quad x \geq 0, \qquad y \geq 0.$$

Method. Treat both non-negativity restrictions as explicit $g_j(x, y) \geq 0$ constraints. Use complementary slackness to test the active lower-bound branch.

Solution. Define

$$g_1(x, y) = x, \qquad g_2(x, y) = y.$$

The Lagrangian is

$$\mathcal{L}(x, y, \lambda_1, \lambda_2) = -(x+2)^2 - 2(y-1)^2 + \lambda_1 x + \lambda_2 y.$$

The complete conditions are

$$-2(x+2) + \lambda_1 = 0, \qquad -4(y-1) + \lambda_2 = 0,$$

$$x \geq 0, \qquad y \geq 0, \qquad \lambda_1 \geq 0, \qquad \lambda_2 \geq 0,$$

$$\lambda_1 x = 0, \qquad \lambda_2 y = 0.$$

The unconstrained maximiser $(-2, 1)$ violates $x \geq 0$. On the branch $x = 0$ and $y > 0$, complementary slackness gives $\lambda_2 = 0$. Stationarity then gives

$$y = 1, \qquad \lambda_1 = 4.$$

All primal, dual and complementary-slackness conditions hold, so

$$(x^*, y^*) = (0, 1), \qquad (\lambda_1^*, \lambda_2^*) = (4, 0).$$

The Hessian is $\mathrm{diag}(-2,-4)$, which is negative definite, and the feasible set is convex. The feasible Kuhn–Tucker point is therefore the unique global maximiser.

Interpretation. The lower bound on x prevents movement towards the unconstrained maximiser. Its positive multiplier is attached to the explicit course constraint $x \geq 0$; the slack lower bound on y has multiplier zero.

Estimator terminology and properties (S7)

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} F_\theta$.

- A **parameter** θ indexes the population model and is fixed but unknown in repeated sampling.
- An **estimand** is the population quantity to be learned. It may be θ itself or a function such as θ^2.
- An **estimator** $\hat{\theta} = T(X_1, \ldots, X_n)$ is a random variable before the sample is observed.
- An **estimate** is the realised numerical value $T(x_1, \ldots, x_n)$.

The bias, variance and mean squared error are

$$\mathrm{Bias}_\theta(\hat{\theta}) = \mathbb{E}_\theta[\hat{\theta}] - \theta,$$

$$\mathrm{Var}_\theta(\hat{\theta}),$$

and

$$\begin{aligned} \mathrm{MSE}_\theta(\hat{\theta}) &= \mathbb{E}_\theta[(\hat{\theta} - \theta)^2] \\ &= \mathrm{Var}_\theta(\hat{\theta}) + \mathrm{Bias}_\theta(\hat{\theta})^2. \end{aligned}$$

An estimator sequence $\hat{\theta}_n$ is consistent for θ if, for every $\varepsilon > 0$,

$$\mathbb{P}_\theta\left(|\hat{\theta}_n - \theta| > \varepsilon\right) \longrightarrow 0 \quad \text{as } n \to \infty.$$

Unbiasedness is a finite-sample expectation statement; consistency is a large-sample probability statement. Neither property is determined by one observed estimate.

Example 3 — two unbiased estimators with different consistency (S7)

Let

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{Bernoulli}(p), \qquad 0 < p < 1.$$

Compare

$$\hat{p}_{1,n} = X_1 \qquad \text{and} \qquad \hat{p}_{2,n} = \bar{X}.$$

Method. Compute expectation, variance and mean squared error, then apply the definition of consistency.

Solution. Both estimators are unbiased:

$$\mathbb{E}_p[\hat{p}_{1,n}] = \mathbb{E}_p[X_1] = p$$

and

$$\mathbb{E}_p[\hat{p}_{2,n}] = \mathbb{E}_p[\bar{X}] = p.$$

Their variances, and therefore their mean squared errors, are

$$\mathrm{MSE}_p(\hat{p}_{1,n}) = \mathrm{Var}_p(X_1) = p(1-p)$$

and

$$\mathrm{MSE}_p(\hat{p}_{2,n}) = \mathrm{Var}_p(\bar{X}) = \frac{p(1-p)}{n}.$$

The second mean squared error tends to zero. For every $\varepsilon > 0$, Chebyshev's inequality gives

$$\mathbb{P}_p(|\bar{X} - p| > \varepsilon) \leq \frac{p(1-p)}{n\varepsilon^2} \longrightarrow 0.$$

Thus $\hat{p}_{2,n}$ is consistent.

By contrast, choose $0 < \varepsilon < \min\{p, 1-p\}$. Since X_1 is either zero or one,

$$|X_1 - p| > \varepsilon$$

with probability one. Therefore $\hat{p}_{1,n} = X_1$ is not consistent, even though it is unbiased.

Interpretation. Increasing n improves the sample mean because it uses the additional observations. The estimator X_1 ignores them, so its sampling uncertainty does not shrink.

Method of moments and maximum likelihood (S7)

The method of moments matches population moments to their sample counterparts. For a one-parameter model, use the lowest raw moment that identifies the parameter. When the finite mean $m(\theta) = \mathbb{E}_\theta[X]$ identifies θ, solve

$$m(\hat{\theta}_{\mathrm{MOM}}) = \bar{X}.$$

If the mean does not identify θ, a higher raw moment may do so. For example, a second-moment equation matches $\mathbb{E}_\theta[X^2]$ to $n^{-1}\sum_i X_i^2$.

Maximum likelihood treats the observed data $\mathbf{x} = (x_1, \ldots, x_n)$ as fixed and maximises

$$L(\theta; \mathbf{x})$$

over the stated parameter space. A score equation identifies an interior candidate, not automatically the global maximum. Check curvature, monotonicity or boundary behaviour as appropriate.

Example 4 — moments and likelihood for a normal variance (S7)

Let

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(0, \theta), \qquad \theta > 0.$$

Derive the method-of-moments and maximum-likelihood estimators of the variance θ. Treat separately the case in which $\sum_i x_i^2 = 0$.

Method. The first raw moment is zero for every θ, so use $\mathbb{E}_\theta[X^2] = \theta$ for the moment equation. For maximum likelihood, write $S_2 = \sum_i x_i^2$, form the joint density, and check the score sign and both parameter-space boundaries.

Solution. Matching the second raw moment gives

$$\hat{\theta}_{\text{MOM}} = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

For any observed real values, the likelihood is

$$L(\theta; \mathbf{x}) = (2\pi\theta)^{-n/2} \exp\left(-\frac{S_2}{2\theta}\right), \qquad \theta > 0.$$

After removing terms that do not depend on θ, the log-likelihood is

$$\ell(\theta; \mathbf{x}) = -\frac{n}{2}\log\theta - \frac{S_2}{2\theta} + \text{constant}.$$

When $S_2 > 0$, the score is

$$\frac{\partial \ell}{\partial \theta} = -\frac{n}{2\theta} + \frac{S_2}{2\theta^2} = \frac{S_2 - n\theta}{2\theta^2}.$$

It is positive below S_2/n, zero at S_2/n, and negative above S_2/n. Moreover, the log-likelihood tends to $-\infty$ as $\theta \downarrow 0$ and as $\theta \to \infty$. Therefore the unique global maximum is

$$\hat{\theta}_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

If $S_2 = 0$, however, the likelihood is proportional to $\theta^{-n/2}$ and is unbounded as $\theta \downarrow 0$. No finite MLE exists on $\theta > 0$. The second-moment equation gives zero, which is outside the parameter space, so there is also no admissible finite method-of-moments estimate in this case.

Interpretation. On positive-squared-sum samples, the two methods happen to give the sample second raw moment in this normal variance model. This is not the sample mean used in a Poisson mean problem, and agreement between the two methods remains model-specific.

Reproduction check

Before starting the exercise sheet, reproduce:

1. the nonlinear equality-constrained candidate comparison and envelope check in Example 1;
2. all four Kuhn–Tucker condition groups and the active lower-bound solution in Example 2;
3. the two variance calculations and consistency argument in Example 3; and
4. the second-moment and normal-variance likelihood derivations, including the zero-squared-sum case, in Example 4.

If you cannot state the constraint encoding, parameter space or regularity condition that makes a step valid, return to that step before memorising the final formula.